

convolution
entropy
see note

set 7

} formula

T → two-headed coin
(not tail here)

1) $P(F|HH)$

$$P(HH|F) \cdot \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

If coin is two-headed

$$P(HH|T) = 1$$

$$P(HH) = P(HH|F)P(F) + P(HH|T)P(T)$$

$$= \frac{1}{4} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2}$$

$$= \frac{5}{8}$$

$$P(F|HH) = \frac{P(HH|F) \cdot P(F)}{P(HH)}$$

$$= \frac{\frac{1}{4} \cdot \frac{1}{2}}{\frac{5}{8}}$$

$$= \frac{1}{5}$$

Intuition, T + 32

40 experiments

case 1: 20 times fair coin

5 is HHH

$$\frac{20}{4} = 5$$

Case 2: 20 times 2 heads

all 20 HH

$$P(\text{Fair} | \text{HH}) = \frac{5}{5+20} = \frac{5}{25} = \frac{1}{5}$$

$$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

$$\frac{P}{8}$$

$$P(\text{HH} | \text{HH}) = P(\text{HH} | \text{HH})$$

$$P(\text{HH})$$